

30 participants and expanded to 1,000 valid responses. Data were cleaned and analyzed using multivariate regression via SPSS. Simultaneously, semi-structured interviews were conducted, and qualitative data were coded and analyzed with NVivo, enhancing the depth and validity of findings. This integrative approach provides a comprehensive understanding of how digital technologies influence personal health behaviors.

Findings: Digital intelligence technology positively impacts health by promoting knowledge dissemination, physical activity, nutritional control, and overall well-being. Analysis reveals a strong correlation between its use and improved health outcomes. However, challenges remain, particularly regarding data confidentiality and equitable access to digital resources. Addressing these issues requires both academic rigor and thoughtful, ethical consideration to ensure technology benefits are accessible, secure, and sustainable for all users.

Conclusion: Digital intelligence technology is reshaping healthy lifestyles with personalized, smart solutions, while also introducing challenges such as data security, privacy concerns, digital inequality, and potential new health risks that require thoughtful and ethical consideration.

<https://doi.org/10.1016/j.amepre.2025.107884>

Cabergoline Impairs Neural Network Efficiency: EEG-Based Evidence from Machine Learning Analysis and Preventive Implications for Cognitive Decline

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Objective: This study investigates the effects of the D2 receptor agonist cabergoline on neural activity associated with Visual-Spatial Working Memory (VSWM), aiming to enhance understanding of pharmacological modulation of cognitive functions and explore its potential implications for preventive medicine, particularly in cognitive health preservation.

Subjects and Methods: This study uses the OpenNeuro public dataset, which includes data from 27 healthy adults who received either cabergoline or a placebo. Electroencephalography (EEG) data were analyzed to extract key neural network features, such as Power Spectral Density (PSD), Phase Locking Value (PLV), and small-world network properties. Machine learning

techniques, including Recursive Feature Elimination (RFE) and random forest classifiers, were applied to assess the impact of cabergoline on neural efficiency and synchronization.

Results: The results indicate significant changes in neural signal synchronicity and brain network efficiency following cabergoline administration, with the PLV matrix showing the most notable effect. The drug appears to disrupt neural synchronization, potentially impairing VSWM capacity. The random forest model achieved 99% test accuracy, indicating robust classification of the drug's effects on cognitive performance.

Conclusions: This study provides new insights into the neural mechanisms of D2 receptor agonists like cabergoline, revealing potential implications for cognitive health. These findings contribute to the development of cognitive health management strategies, offering valuable guidance for early detection and intervention in cognitive decline, particularly in aging or neurodegenerative conditions, with broader applications in preventive medicine.

<https://doi.org/10.1016/j.amepre.2025.107885>

Optimizing Aerobics Club Environments to Enhance Women's Health and Exercise Adherence: The Mediating Role of Perceived Value

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Objectives: This study examines the mediating effect of perceived value on the relationship between the service scape of aerobics clubs and women's intention to continue participating in aerobics. As the global focus on healthy living intensifies, aerobics has gained popularity among women due to its cardiovascular benefits, mental health improvements, and opportunities for social engagement. However, the mechanisms through which an optimized club environment enhances perceived value and fosters long-term exercise adherence remain underexplored. This study aims to bridge this gap and provide strategic insights for club operators to promote sustainable health development.

Subjects and Methods: A survey was conducted among female aerobics club members using a convenience sampling method. A total of 314 valid responses were collected through online and offline surveys. The survey included demographic characteristics, service scape, perceived value, and aerobics continuation intention, with items measured on a 5-point Likert scale. Exploratory